

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 7

1. Confidence Interval

- Task: given i.i.d. random sample $X_1, \dots, X_n$ from unknown distribution P and confidence level $1 - \alpha$, we hope to find interval $I = [a(X_1, \dots, X_n), b(X_1, \dots, X_n)]$ with $a(X_1, \dots, X_n), b(X_1, \dots, X_n) \in \mathbb{R} \cup \{-\infty, +\infty\}$ such that

$$\mathbb{P}(\mu \in I) = 1 - \alpha, \quad (1)$$

where $\mu = \mathbb{E}_{X \sim P}[X]$ is the population mean. We call I as $100(1 - \alpha)\%$ confidence interval (CI). e.g. α Here are some remarks

- (a) The interpretation of (1): it provide theoretical guarantees on mistake we can make. If we repeat the experiments T times, then $(1 - \alpha)T$ of the interval we constructed covers μ . In other words, we can only make mistakes (I doesn't cover μ) among approximately $100\alpha\%$ of the experiments).
- (b) α control the tolerance of mistake. Intuitively, when α decreases, the length of the CI increase.
 - Extreme case: $\alpha = 0$, we can not make mistake, the CI should be $(-\infty, \infty)$ if X is unbounded.
- (c) $\bar{X}$ is a good estimate of μ (by LLN), its natural to construct the CI as
 - two-side CI: $[\bar{X} - \delta, \bar{X} + \delta]$
 - one-side CI: (upper bound CI) $(-\infty, \bar{X} + \delta]$; (lower bound CI) $[\bar{X} - \delta, \infty)$.
- (d) Why two-side CI looks like $[\bar{X} - \delta, \bar{X} + \delta]$ rather than, for example, $[\bar{X} - 2\delta, \bar{X} + \delta]$.
 - By CLT, $\sqrt{n}(\bar{X} - \mu)/\sigma \approx \mathcal{N}(0, 1)$. It is symmetric, its PDF value $f(x)$ decreases when x increases in $x \in [0, \infty)$.
 - The choice of symmetric CI allows the width of CI to be minimal for fixed tolerance parameter α . Formally, the following optimization objective

$$\min_{a,b} (b - a) \quad s.t. \quad \mathbb{P}(a \leq Z \leq b) \geq 1 - \alpha$$

attains minimal value when $b = -a = z_{\alpha/2}$.

- Suppose we **know the variance** σ^2 of $\mathcal{P}$.

i) If $\mathcal{P} \sim \mathcal{N}(\mu, \sigma^2)$

- We have $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$ exactly.
- This helps us construct confidence intervals of the form $\bar{X} \pm \frac{\sigma}{\sqrt{n}}\delta$, $(-\infty, \bar{X} + \frac{\sigma}{\sqrt{n}}\delta]$, $[\bar{X} - \frac{\sigma}{\sqrt{n}}\delta, \infty)$.
 - * For two-side $(1 - \alpha)$ -CI

$$\begin{aligned} \mathbb{P}\left[\bar{X} - \delta \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + \delta \frac{\sigma}{\sqrt{n}}\right] &= 1 - \alpha \iff \mathbb{P}\left[-\delta \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \delta\right] = 1 - \alpha \\ &\iff \Phi(\delta) - \Phi(-\delta) = 1 - \alpha \\ &\iff \alpha/2 = 1 - \Phi(\delta) \end{aligned}$$

width $= 2z_{\alpha/2}\sigma/n$, increases as α decreases (higher confidence-level) and decreases as n increases.

- * For the one-side lower bound

$$\mathbb{P}\left[\bar{X} - \delta \frac{\sigma}{\sqrt{n}} \leq \mu\right] = 1 - \alpha \iff \mathbb{P}\left[\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq \delta\right] = 1 - \alpha \iff \Phi(\delta) = 1 - \alpha \iff \alpha = 1 - \Phi(\delta).$$

- Two side $(1-\alpha)$ CI: $\delta = z_{\alpha/2}$; one side $(1-\alpha)$ CI: $\delta = z_{\alpha}$. Here critical value z_{α} satisfies $\mathbb{P}(Z > z_{\alpha}) = \alpha$.
- ii) If $\mathcal{P}$ is an unknown distribution and sample size is sufficiently large ($n \geq 30$).
 - By CLT $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ is approximately $\mathcal{N}(0, 1)$
 - This is an approximation (but a very good one)
- Suppose that we **do not know** the variance σ^2 of $\mathcal{P}$. We estimate σ^2 by the sample variance $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.
 - i) If $\mathcal{P}$ is an unknown distribution and the sample size is sufficiently large ($n \geq 40$):
 - By CLT $\sqrt{n}(\bar{X} - \mu)$ is approximately $N(0, \sigma^2)$. By LLN $s^2 \approx \sigma^2$. $\sqrt{n}(\bar{X} - \mu)/\sqrt{s^2} \approx \mathcal{N}(0, 1)$
 - Again, this is an approximation (but also tends to be very good)
 - ii) If $\mathcal{P} \sim \mathcal{N}(\mu, \sigma^2)$, sample size is small.
 - We have $\sqrt{n}(\bar{X} - \mu)/\sqrt{s^2} \sim t_{n-1}$, student- t distribution with degree of freedom $n - 1$ exactly.
 - A few remarks on student-t distribution and the implications on CI construct in this case.
 - (a) Visualization of PDFs for $\mathcal{N}(0, 1)$ (t_{∞}), t_2 and t_{10} .
 - (b) student-t distribution is heavy-tailed, we have $t_{m,\alpha} > z_{\alpha}$.
 - * For example, $z_{0.025} \approx 1.96$, $t_{10,0.025} \approx 2.22$, $t_{40,0.025} \approx 2.00$, $t_{100,0.025} \approx 1.98$.
 - * It implies the CI in this case is wider than that constructed above.
 - (c) For fixed α , $t_{m,\alpha}$ decreases as m increases and $\lim_{m \rightarrow \infty} t_{m,\alpha} = z_{\alpha}$.
 - * It implies the gap between the CI in this case and that in i) is getting smaller and are approximately the same when n is large enough.

2. A summary of methods

$\mathcal{P}$	two-sided C.I	upper C.I	lower C.I
$\mathcal{N}(\mu, \sigma^2)$	$[\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}]$	$(-\infty, \bar{x} + z_{\alpha} \frac{\sigma}{\sqrt{n}}]$	$[\bar{x} - z_{\alpha} \frac{\sigma}{\sqrt{n}}, \infty)$
σ^2 known, n large			
σ^2 unknown, n large	$[\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}}]$	$(-\infty, \bar{x} + z_{\alpha} \frac{s}{\sqrt{n}}]$	$[\bar{x} - z_{\alpha} \frac{s}{\sqrt{n}}, \infty)$
σ^2 unknown, n small, $\mathcal{N}(\mu, \sigma^2)$	$[\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}]$	$(-\infty, \bar{x} + t_{\alpha, n-1} \frac{s}{\sqrt{n}}]$	$[\bar{x} - t_{\alpha, n-1} \frac{s}{\sqrt{n}}, \infty)$

3. Examples

- (a) An article gave the following summary information for fracture strengths (in MPa) of 169 ceramic bars. The average is 89.1 MPa.
 - i. Find the 95% confidence interval for average fracture strength if the sample standard deviation is 3.73.
 - ii. Suppose the population standard deviation is $\sigma = 4$. How large does the sample size need to be to obtain a 95% C.I. for μ with the width of at most 0.5?
- (b) Suppose a simple random sample of 150 students is drawn from a population of 3000 college students. Among sampled students, the average IQ score is 115 with a standard deviation of 10. What is the 99% confidence interval for the students' IQ score?